Facial Recognition Using OpenCV

Question: In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.

Explanation

Image acquisition captures the input image. Low-level vision preprocesses the image using grayscale conversion, Gaussian blur, and histogram equalization. Mid-level vision detects the face using the Haar Cascade classifier. High-level vision makes the final decision based on whether a face is detected.

Complete Python Code

```
import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Step 1: Upload Image
uploaded = files.upload()
image_path = list(uploaded.keys())[0]
image = cv2.imread(image_path)

# Step 2: Preprocessing (Low-Level Vision)
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
gray = cv2.GaussianBlur(gray, (5,5), 0)
gray = cv2.equalizeHist(gray)

# Step 3: Face Detection (Mid-Level Vision)
face_detector = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml")

faces = face_detector.detectMultiScale(
    gray,
    scaleFactor=1.1,
    minNeighbors=5,
    minSize=(30,30)
)

rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

for (x,y,w,h) in faces:
    cv2.rectangle(rgb, (x,y), (x+w,y+h), (0,255,0), 2)

plt.imshow(rgb)
plt.title("Detected Face")
plt.axis("off")
plt.show()

# Step 4: High-Level Vision (Decision)
if len(faces) > 0:
    print("Face Detected")
else:
    print("No Face Detected")

print("Number of Faces:", len(faces))
```